Step Into Computer Science

Overview: This year-long course equips students with the foundational knowledge of computational thinking and computer science. Through interactive exercises using block-based programming, students will delve into the exciting world of coding, where they will create animations, games, art, and stories, fostering creativity and problem-solving skills.

Throughout the course, participants will gain a solid understanding of the basics of programming, laying the groundwork for more advanced studies in computer science. Emphasis will be placed on applying computational thinking to real-life situations, allowing students to see the practical applications of the concepts they learn.

Objectives:

- Develop problem solving skills while learning how computers process information to solve problems. Know the basic parts and functions of the computer.
- Create and share content on individual web pages while exploring the impact of sharing on-line information and learning to think thinking critically about on-line content. Learn the basic differences between spreadsheets, documents, presentations, and image usage while developing their webpage.
- Create images, animation, art, stories, and games while developing computational thinking skills and exploring programming concepts and the design process computer scientists use on a daily basis.
- Consider the social impacts of computing as they develop solutions to needs in the community using the design process.
- Discover the importance of using data to solve problems and the way computers aid in this process.
- Explore the role of physical devices in computing while creating programs that utilize micro:bits and developing prototypes using the design process.
- Develop a machine learning projects and apps using real world data and solving community problems.

Assessments: Project rubrics, pre-post test, mini-quizzes and/ or exit tickets

Course Essentials:

Equipment	**Cost/Unit**
	TBD
Computer or laptop	
Other Materials	Reusable: $1000 Consumable: (up to $300 per year, replace as needed)

First Semester Course Outline

Unit 1: Problem-solving	Introduction to problem solving and computer science. Using various challenges to learn how to problem solve effectively. Explore what is a computer and how do they help people solve problems. Consider online safety and responsible use of technology. Project.
Unit 2: Web page development and common computer applications	Explore the role of the web, and the impact of information sharing on-line. Learn how to be a more critical on-line content consumers and consider intellectual property. Skill building with debugging. HTML, CSS, image usage, spreadsheets, presentations, and documents. Project

Unit 3: Interactive animations and games	Interactive animation and game development. Investigate animation and game design, coordinate systems, shapes, using sprites, and design process while exploring computational thinking and programming skills such as debugging, sequencing, variables, randomness, conditionals, inputs, functions, etc.

Second Semester Course Outline

Unit 4: Data	What is data, and how do computers represent data. Students use data sets to solve problems. Explore patterns, binary and ASCII representation. Understand the importance of data privacy. Students experiment with making decisions with data and computers, including big data and machine learning.
Unit 5: Physical computing and the design process	Students explore the design process. Research and Understanding the users needs and design constraints, prototype, testing and debugging, feedback. Teamwork and collaboration practices. App lab usage explored. Javascript. Micro:bit parts and functions. Makecode and extensions. Physical inputs and outputs, sensors, variables, algorithms and conditionals. Collecting data.
Unit 6: Machine learning and AI	What is artificial intelligence? What is machine learning? Using the design process to train a computer to solve problems. Metal models and numerical models are explained and explored. Patterns in data discussed. Algorithms ad classification. Model bias is explored, and the ethics for using AI. Project.

STEP INTO COMPUTER SCIENCE (MIDDLE SCHOOL ELECTIVE)

1. Materials
   Internet access, one-to-one computer use daily, and access to the LSU servers. All materials and activities are accessible via Chromebooks and iPads or Tablets.

Hardware	Qty Needed	Price
BBC micro:bit BBC Micro:Bit V2.1 Go Kit - Includes micro:bit Board MicroUSB Cable, and Battery Pack (1) micro:bit v2 CLUB Bundle (set of 10) Parallax micro:bit Classroom Pack (pack of 30)	Class set/or one per pair of students	1 - $19.95 Pack of 10 - $175 Classroom set - $600
Finch Robots (1, includes micro:bit) Finch Flock (set of 5)	Class set/or one per pair or group of students	Price varies $180 (individual) or $950 (set of 5)
Hummingbird kit* (includes micro:bit) Hummingbird Flock* (set of 4) *only needed if you are choosing Path 2 in Unit 6	Class set/or per pair or group	Price varies $130 (individual, basic) or $920 (set of 4 premium)

A complete list of materials can be found here.

2. Required software, networking access, and access to LSU servers
   - Students will need to sign up with online development and testing environments, including but not limited to code.org, scratch.mit.edu, and others.
   - Students will need access to YouTube instructional videos relevant to the course, as well as other educational video repositories.
   - Teachers will need to be able to access the LSU servers using several Internet protocols including but not limited to HTTPS and SSH.
   - Teachers will need to share sample student work with their designated LSU Pathway Point-of-Contact.
   - Principals will need to communicate with the district's information technology department to ensure that there are no technological restrictions that block access to the LSU servers in the lsu.edu, college-readiness.lsu.edu, louisianatrainingplatform and pathways.lsu.edu domains on any port.
   - A complete whitelist can be found here.

3. Required teacher collaborations. Teachers will communicate with LSU instructors via emails, Google Drive, and/or apps hosted on the LSU servers.

4. Required administration of course content, pre/post test, and research instruments. All required materials and instruments will be either posted in the LSU servers, Google Drive, or their location announced via email.

5. Course Work
   Teachers must present the course material in sequence or as approved by collaboration with the LSU Pathway Point-of-Contact. Teachers are expected to deliver a minimum of 80% of the course material for course fidelity and for the opportunity for IBC attainment (where applicable).

6. Other. As this is a project-based learning class, we strongly suggest that each section of the course be limited to a *maximum* of 25 students. The course is dependent on the teacher providing feedback and reviewing student code. The course requires that teachers have adequate time to interact with each student.